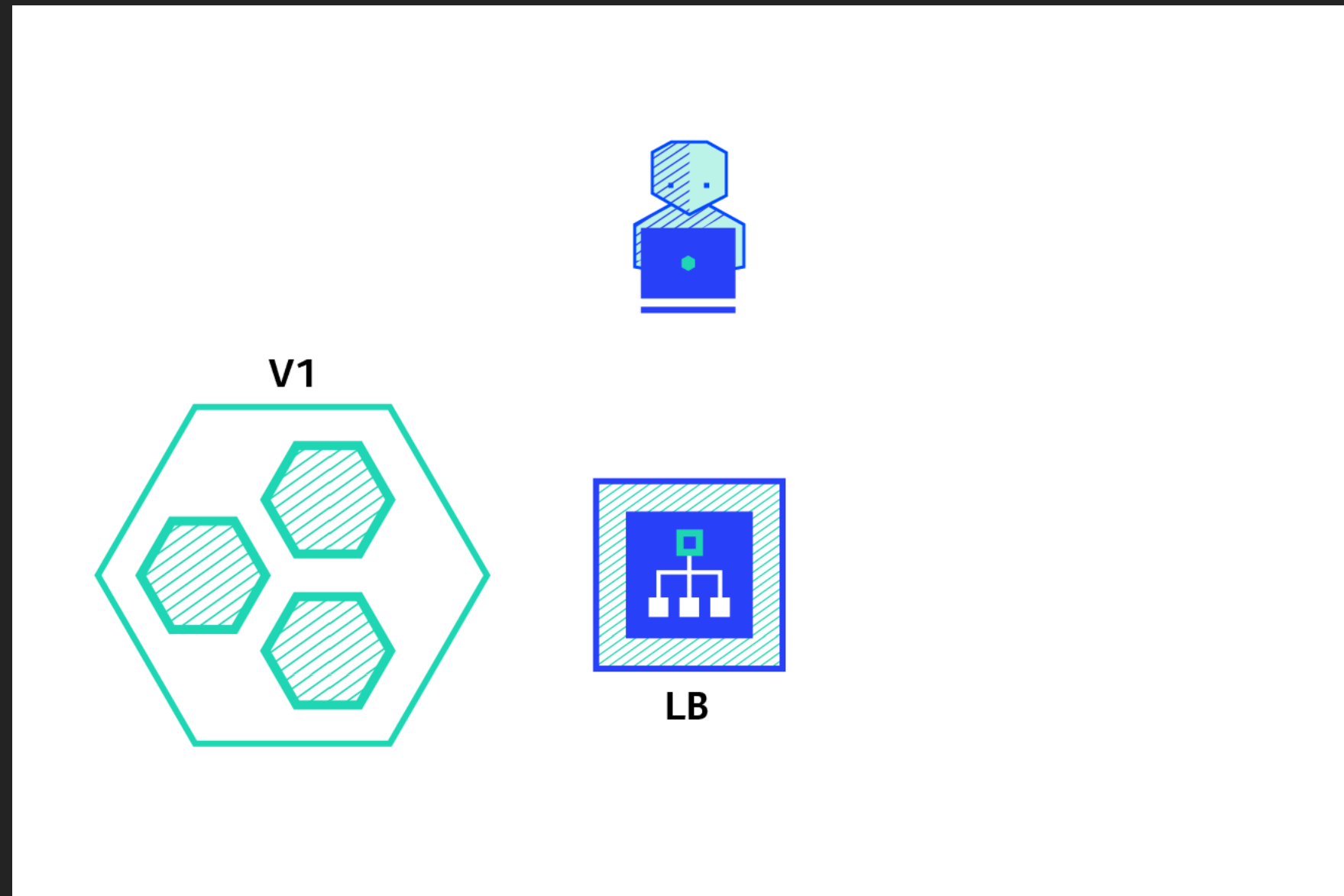


CI/CD with Google Cloud Build

Outline

- » Canary deployment
- » Cloud Deploy configuration
- » Cloud Deploy scenarios
- » Rollout with approval restriction
- » Demo
- » Integrate with current ci

Canary deployment



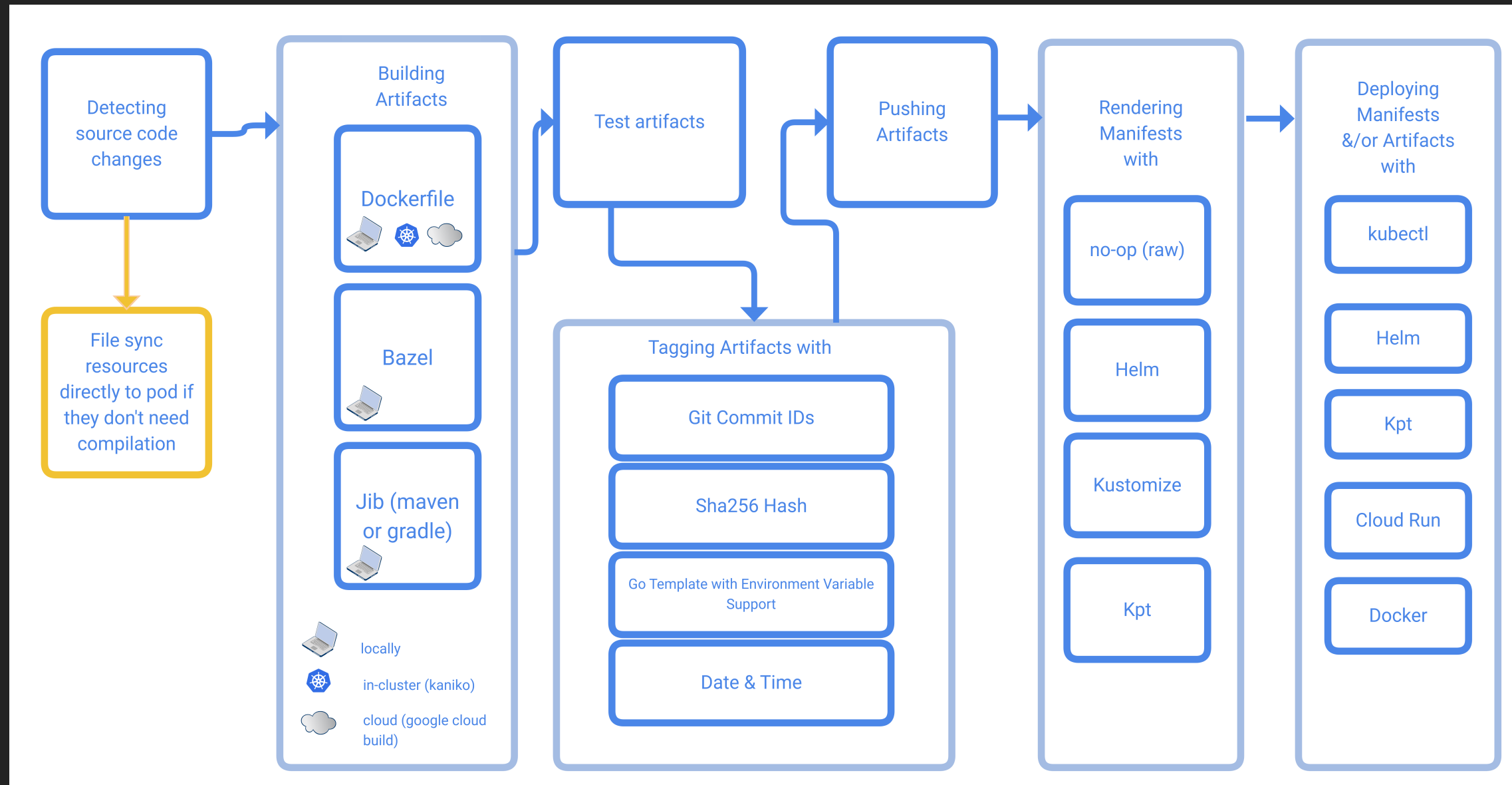
Canary deployment

- » Deploy 2 version app at same time
- » Configure load balancer route small amount of traffic to new version
- » Monitor error rate see if new version is ok to rollout
- » Rollout/Rollback according to monitor result

Cloud deploy configuration

- » skaffold.yaml: Skaffold configuration
- » clouddeploy.yaml: Config **delivery pipeline** and **targets**

Skaffold



skaffold.yaml

```
apiVersion: skaffold/v4beta4
kind: Config
manifests:
  rawYaml:
    - kubernetes.yaml
deploy:
  kubectl: {}
verify:
- name: verification-curl
  container:
    name: verification-curl
    image: curlimages/curl:8.4.0
    command: ["sh", "-c"]
    args: ['response=$(curl -s my-service.default.svc.cluster.local/debug) && (echo "$response" | grep -q "bug") && exit 1 || exit 0']
executionMode:
  kubernetesCluster: {}
```

skaffold.yaml – render manifest ¹

```
apiVersion: skaffold/v4beta4
kind: Config
manifests:
  rawYaml:
    - kubernetes.yaml
deploy:
  kubectl: {}
verify:
- name: verification-curl
  container:
    name: verification-curl
    image: curlimages/curl:8.4.0
    command: ["sh", "-c"]
    args: ['response=$(curl -s my-service.default.svc.cluster.local/debug) && (echo "$response" | grep -q "bug") && exit 1 || exit 0']
  executionMode:
    kubernetesCluster: {}
```

¹[Skaffold render manifest](#)

skaffold.yaml – deploy²

```
apiVersion: skaffold/v4beta4
kind: Config
manifests:
  rawYaml:
    - kubernetes.yaml
deploy:
  kubectl: {}
erify:
- name: verification-curl
  container:
    name: verification-curl
    image: curlimages/curl:8.4.0
    command: ["sh", "-c"]
    args: ['response=$(curl -s my-service.default.svc.cluster.local/debug) && (echo "$response" | grep -q "bug") && exit 1 || exit 0']
executionMode:
  kubernetesCluster: {}
```

²Skaffold deploy

skaffold.yaml – verify³

```
apiVersion: skaffold/v4beta4
kind: Config
manifests:
  rawYaml:
    - kubernetes.yaml
deploy:
  kubectl: {}
verify:
- name: verification-curl
  container:
    name: verification-curl
    image: curlimages/curl:8.4.0
    command: ["sh", "-c"]
    args: ['response=$(curl -s my-service.default.svc.cluster.local/debug) && (echo "$response" | grep -q "bug") && exit 1 || exit 0']
  executionMode:
    kubernetesCluster: {}
```

³Cloud deploy – canary

clouddeploy.yaml

```
apiVersion: deploy.cloud.google.com/v1
kind: DeliveryPipeline
metadata:
  name: my-demo-app-single-region
description: main application pipeline
serialPipeline:
  stages:
  - targetId: cluster-dev
    profiles: []
```

```
---
apiVersion: deploy.cloud.google.com/v1
kind: Target
metadata:
  name: cluster-dev
description: development cluster
gke:
  cluster: projects/swag-nino-chang/locations/us-central1/clusters/quickstart-cluster-qsdev
```

Deploy Scenario

- » Single / Multi region
- » Canary deploy

Single region

```
apiVersion: deploy.cloud.google.com/v1
kind: DeliveryPipeline
metadata:
  name: my-demo-app-single-region
description: main application pipeline
serialPipeline:
  stages:
  - targetId: cluster-dev
    profiles: []
```

```
---
apiVersion: deploy.cloud.google.com/v1
kind: Target
metadata:
  name: cluster-dev
description: development cluster
gke:
  cluster: projects/swag-nino-chang/locations/us-central1/clusters/quickstart-cluster-qsdev
```

Multi region – DeliveryPipeline

```
apiVersion: deploy.cloud.google.com/v1
kind: DeliveryPipeline
metadata:
  name: my-demo-app-multi-region
description: main application pipeline
serialPipeline:
  stages:
  - targetId: cluster-multi
    profiles: []
    deployParameters:
      - values:
          replicaCount: "1"
          matchTargetLabels:
            label1: label1
      - values:
          replicaCount: "2"
          matchTargetLabels:
            label2: label2
```


Multi region – DeliveryPipeline

```
apiVersion: deploy.cloud.google.com/v1
kind: DeliveryPipeline
metadata:
  name: my-demo-app-multi-region
description: main application pipeline
serialPipeline:
  stages:
  - targetId: cluster-multi
    profiles: []
    deployParameters:
      - values:
          replicaCount: "1"
          matchTargetLabels:
            label1: label1
      - values:
          replicaCount: "2"
          matchTargetLabels:
            label2: label2
```

Multi region – Targets

```
apiVersion: deploy.cloud.google.com/v1
kind: Target
metadata:
  name: cluster-multi
description: production clusters
multiTarget:
  targetIds: [cluster-production-a, cluster-production-b]
---
apiVersion: deploy.cloud.google.com/v1
kind: Target
metadata:
  name: cluster-production-a
  labels:
    label1: label1
description: production cluster 1
gke:
  cluster: projects/swag-nino-chang/locations/us-central1/clusters/quickstart-cluster-qsprod
---
apiVersion: deploy.cloud.google.com/v1
kind: Target
metadata:
  name: cluster-production-b
  labels:
    label2: label2
description: production cluster 2
gke:
  cluster: projects/swag-nino-chang/locations/us-west1/clusters/quickstart-cluster-qsprod2
```

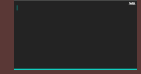

Canary Deploy³

```
apiVersion: deploy.cloud.google.com/v1
kind: DeliveryPipeline
metadata:
  name: my-canary-demo-app-1
description: main application pipeline
serialPipeline:
  stages:
  - targetId: cluster-dev
    profiles: []
    strategy:
      canary:
        runtimeConfig:
          kubernetes:
            serviceNetworking:
              service: "my-service"
              deployment: "my-deployment"
        canaryDeployment:
          percentages: [50]
          verify: true
```

³Cloud deploy - canary

Canary Deploy

```
apiVersion: deploy.cloud.google.com/v1
kind: DeliveryPipeline
metadata:
  name: my-canary-demo-app-1
description: main application pipeline
serialPipeline:
  stages:
  - targetId: cluster-dev
    profiles: []
    strategy:
      canary:
        runtimeConfig:
          kubernetes:
            serviceNetworking:
              service: "my-service"
              deployment: "my-deployment"
        canaryDeployment:
          percentages: [50]
          verify: true
```



Add **requireApproval: true** to enable approve.

```
apiVersion: deploy.cloud.google.com/v1
kind: Target
metadata:
  name: cluster-production
description: production cluster
requireApproval: true
gke:
  cluster: projects/swag-nino-chang/locations/us-central1/clusters/quickstart-cluster-qspod
```

DEMO

Demo Repo

<https://github.com/Robert-Rino/tool-kit/tree/master/cloud-deploy-gke>

Integrate with current ci

Pre-rollout action canary

- » Trigger on master merge, after image build action.
- » Github action create cloud deploy canary release ⁴
- » Create github deployment **canary**
- » Approve request to slack **#dev-ops** channel

⁴ [github action x cloud deploy integration](#)

Integrate with current ci

rollout action

- » Trigger on github webhook
- » Advance release from **canary** to **stable**
- » Remove github deployment **canary**
- » Create github deployment **stable**

Q & A